

Can we trust ACTIONS for Galactic Archaeology?

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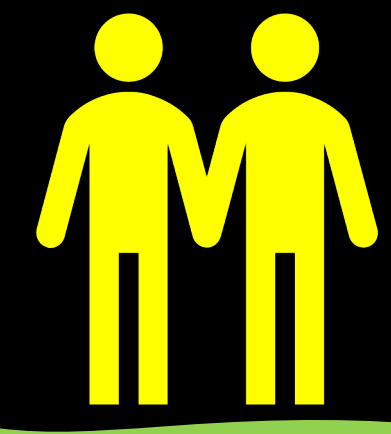
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TAKEAWAY:

Actions are not conserved for a single star over ~100 Myr but stars born together remain clustered in action space for up to 0.5 Gyr. Close-born stars execute a correlated logarithmic random walk in action space.



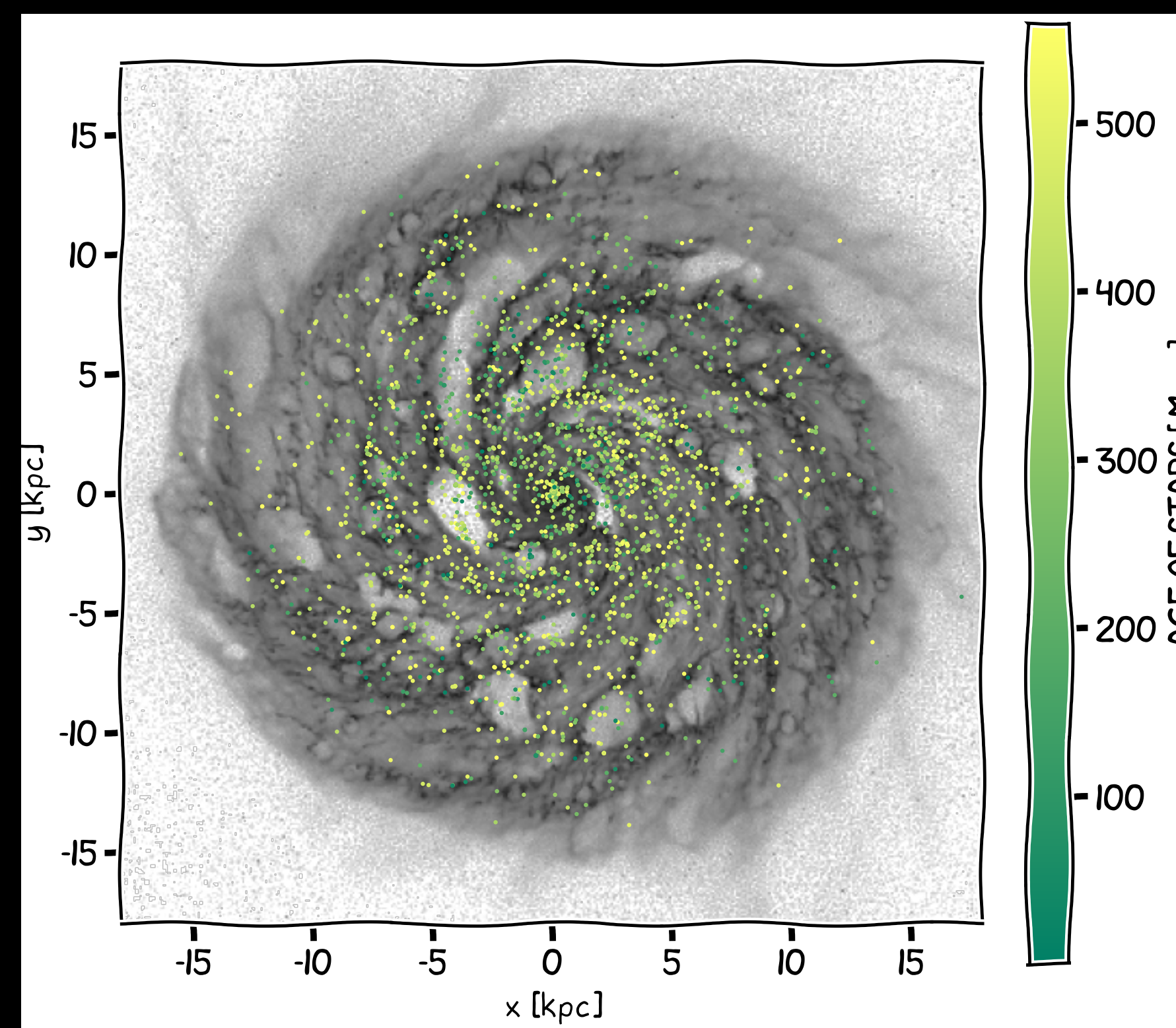
MOTIVATION

Actions (J_R, J_z, J_ϕ) are conserved in steady and axisymmetric potentials. But Milky Way evolves in time and has a bar, spiral arms, etc.

How do actions evolve in the Milky Way?

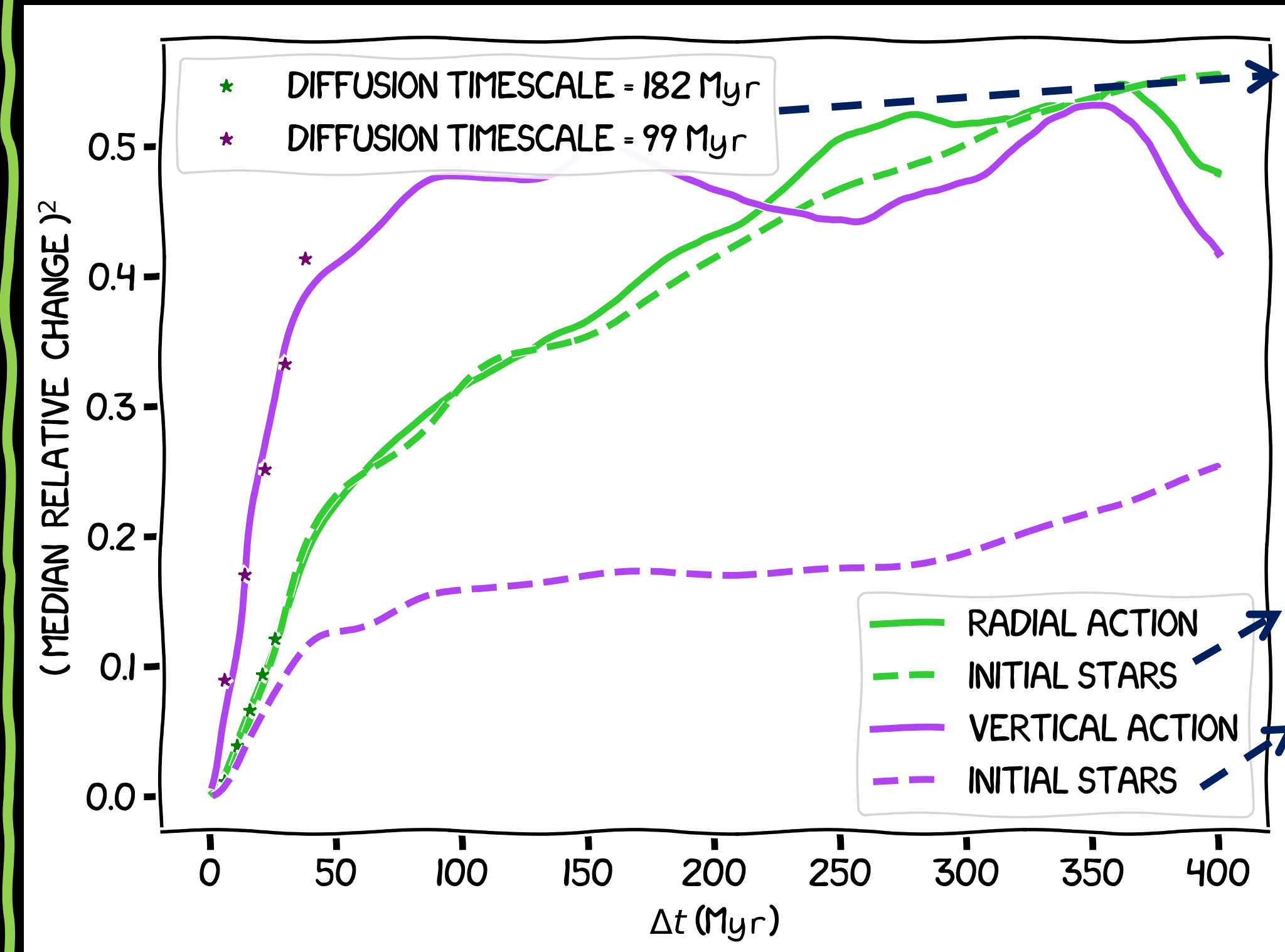
To find out, we use a Milky Way-like simulation with self-gravity, star formation and feedback and:

- 1.32 million star particles
- $286 M_\odot$ mass resolution
- Run for 0.5 Gyr at 1 Myr cadence output



ACTIONS NOT CONSERVED!

- Actions change at ~100 Myr timescales, hence, orbit reconstruction based on assumption of conserved action not possible



Different timescales of random walk for vertical and radial actions.

Stars located much above the midplane show different vertical action change but same radial action change.

Do they have different mechanisms driving the change?

VERTICAL ACTION CHANGE DRIVEN BY GMCs; RADIAL AND AZIMUTHAL ACTION CHANGE BY SPIRAL ARMS!

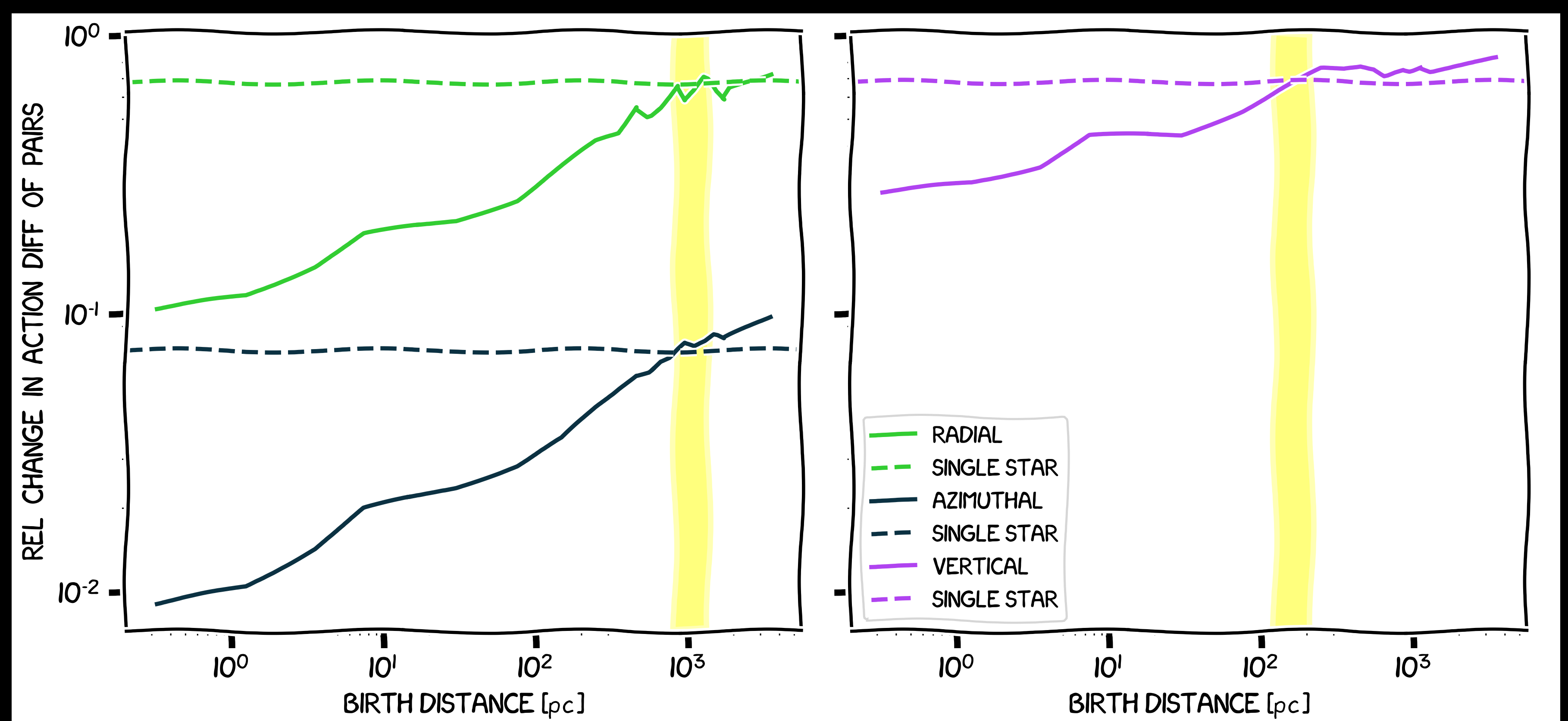
We measure action coherence for stellar pairs by taking the median over time and comparing the change in pair-action difference with the pair's birth separation.

Close-born pairs stay highly clustered in action space: their relative change in difference of actions is very small: HOPE for Galactic Archaeology!

Now the key question is:

At what distance does their evolution become indistinguishable from single-star drift?

It's different for different components!

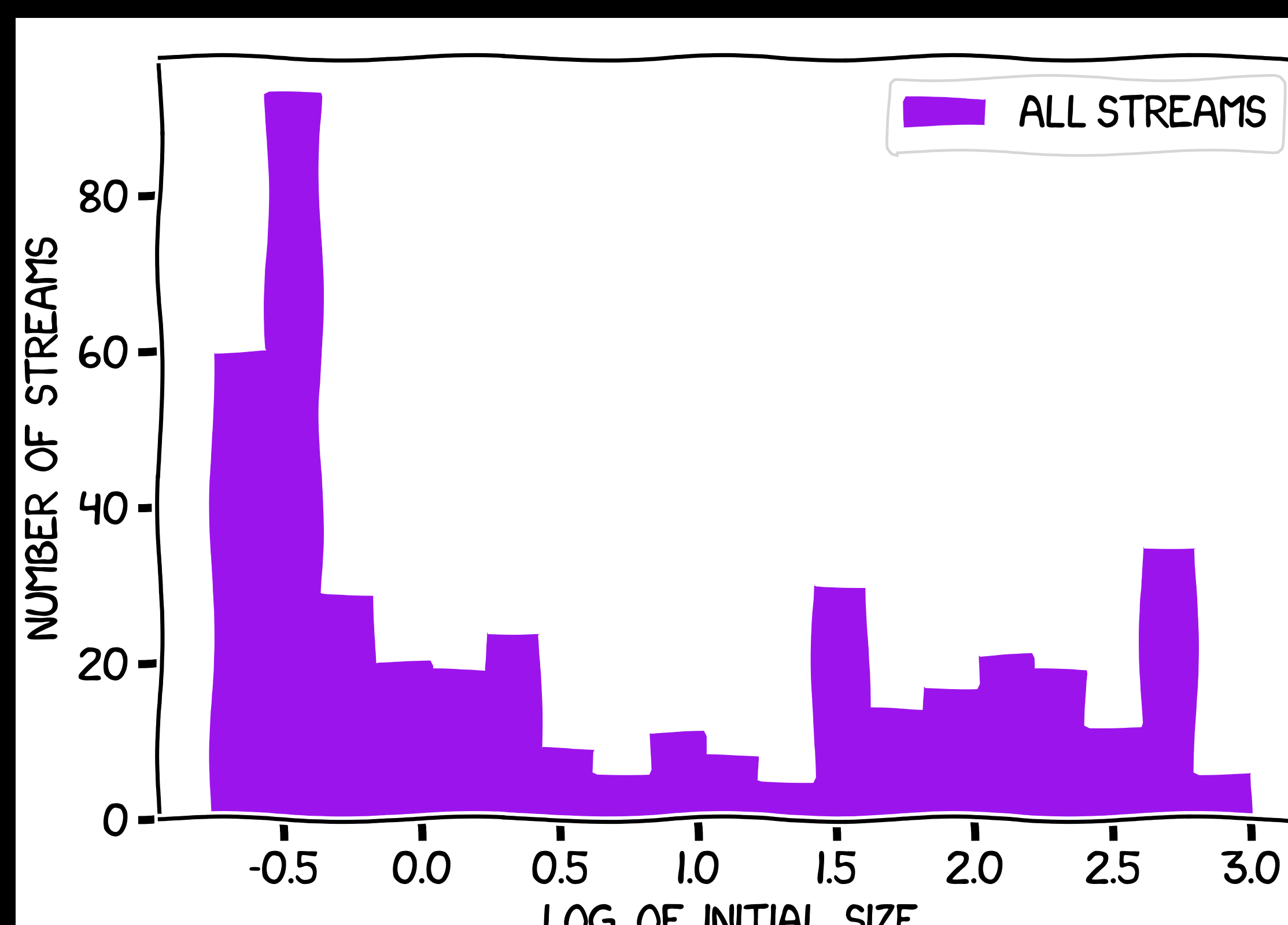
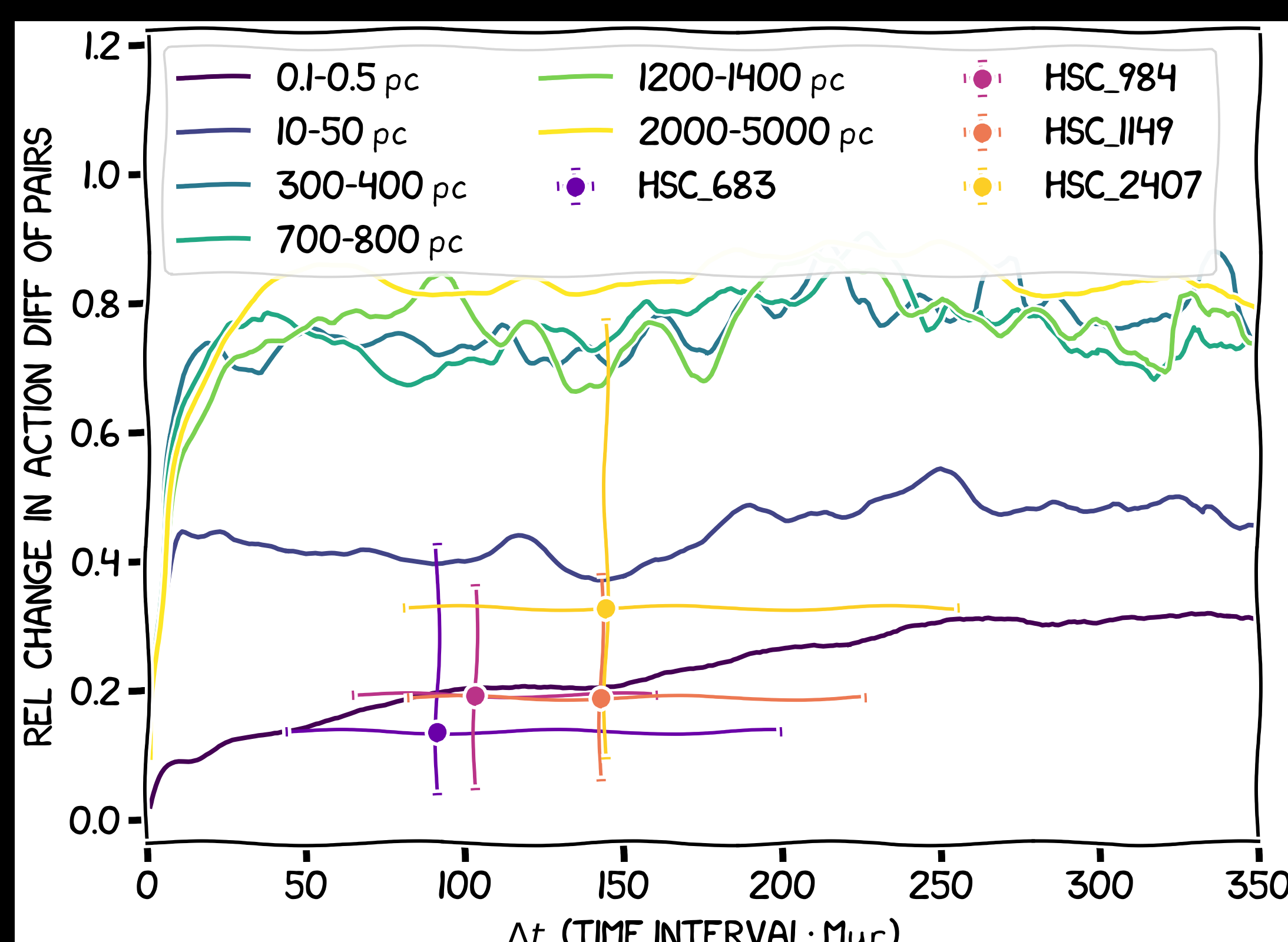


Radial and azimuthal actions decohere at length scales of 1 kpc pointing to spiral arm structures as drivers of change.

However, vertical action decohere at length scale of 100 pc (disc scale height) which points to giant molecular clouds as the drivers of change.

CAN WE USE ACTIONS FOR GALACTIC ARCHAEOLOGY? APPLICATION TO DISC STREAMS

Since stars born together remain clustered in action space, we can ask whether present-day disc streams were born as clusters. We compare the observed actions and ages of 438 disc streams with the changes expected from the simulation to constrain their initial sizes.



If the initial size is small (around <30 pc), the stream is likely a cluster remnant.

Larger inferred initial sizes (upto 1000 pc, right side of the plot) may instead be resonant structures that are kinematically consistent but not born together.

REFERENCES

TALK TO ME @ arunima.arunima@anu.edu.au !!